

Question 1

Zip Codes: 00600-98000 inclusive

Equivalence Partitions:

- value < 00600 (invalid input)
- 00600 <= value <= 98000 (valid input)
- value < 0 (invalid input)
- value > 98000 (invalid input)

Partition Type	Partition Description	Test Cases
invalid	value < 00600	30, 800
valid	00600 <= value <= 98000	00600, 40000
invalid	value < 0	-1, -500
invalid	value > 98000	100000, 101000

States: AL-WY

Equivalence Partitions:

- AL <= value <= WY (valid input)
- value != state (invalid input)
- value < 2 letters (invalid input)
- value > 2 letters (invalid input)

Partition Type	Partition Description	Test Cases
valid	AL <= value <= WY	CA, DE
invalid	value != state	ZZ, LM
invalid	value < 2 letters	M, P
invalid	value > 2 letters	MOP, PAN

Last Name: One through twelve characters (alphabetic, space and hyphen)

Equivalence Partitions:

- value includes number (invalid input)
- value < 1 length (invalid input)
- value > 12 length (invalid input)
- 1 <= value <= 12 length, only letters/spaces/hyphens (valid input)

Partition Type	Partition Description	Test Cases
invalid	value includes number	add4, alij1
invalid	value < 1 length	
invalid	value > 12 length	qwertyuiopadf
valid	1 <= value <= 12 length, only letters/spaces/hyphens	hello world

User ID: Eight characters, at least two of which are not alphabetic (numeric, special characters)

Equivalence Partitions:

- value > 8 length (invalid input)
- value < 8 length (invalid input)
- value contains < 2 non-alphabetic chars (invalid input)
- value = 8 length, has >= 2 non-alphabetic chars

Partition Type	Partition Description	Test Cases
invalid	value > 8 length	asd12fgji
invalid	value < 8 length	12ef
invalid	value has < 2 non-alphabetic chars	1abdhgir
valid	value = 8 length, has >= 2 non-alphabetic chars	12erty3o, 12fjahwu

Question 2

Equivalent Partitions:

- Between 90 and 100, inclusive (valid input)
- Between 80 and 89, inclusive (valid input)
- Between 70 and 79, inclusive (valid input)
- Between 60 and 69, inclusive (valid input)

- 59 and below (valid input)
- Above 100 (invalid input)
- Below 0 (invalid input)

The test cases given below use the boundaries of n-1, n, n+1

Test Case #	Input	Expected Output
1	101	Error
2	100	A
3	99	A
4	91	A
5	90	A
6	89	B
7	88	B
8	81	B
9	80	B
10	79	C
11	78	C
11	70	C
12	69	D
13	68	D
14	61	D
15	60	D
16	59	F
17	58	F
18	1	F
19	0	F
20	-1	F